Sliding Control	이재운	2024. 4. 12

7. Introduction

Control of "imprecise" nonlinear model. Imprecision comes from

1. Actual uncertainty about plant (ex) unknown parameter
2. Simplification of model (ex) linearizing friction, neglecting structural modes in rigid body

In control p.o.v

1. Structured uncertainties : uncertainty on the term. (ex) measure error
2. unstructured uncertainties : " by estimation. (ex) linear approximation

- Uncertainties may have strong adverse effect
  → 2 major & complementary approaches

1) Robust control (Chap 7)

nominal part + uncertainty dealing part

one of simple approaches is

2) Adaptive control (Chap 8)

similar, but updating controller based on performance

Similar to chap 6. Fluid in the tank

$A(h)\dot{h} = u - a\sqrt{2gh} \Rightarrow u = a\sqrt{2gh} + A(h)v$

- Sliding control methodology : n^{th} order system → 1^{st} order systems
  can be solved perfectly

Subjects

7.1 main concepts, notations, basic controller design

7.2 modification of control laws - eliminate excessive control

7.3 choice of controller design parameters

7.4 generalization to multi-input systems

7.1. Introduction

$$x^{(n)} = f(\mathbf{x}) + b(\mathbf{x})u \qquad x \in \mathbb{R},\ u \in \mathbb{R} \qquad \mathbf{x} = [x, \dot{x}, \dots, x^{(n-1)}]^T \in \mathbb{R}^n$$

our interest (x); control input (u)

f, b are not known exactly, but

i) $\exists\ \hat{f}, \underbrace{F}_{\text{cont}}$ s.t $|f - \hat{f}| \le F$ is known (all functions of $\mathbf{x}$)

ii) $\exists\ \hat{b}, \underbrace{B}_{\text{cont}}$ s.t ① sign of b is known, $|b - \hat{b}| \le B$ (")

<Objective>

tracking $\mathbf{x}_d = [x_d, \dot{x}_d, \cdots x_d^{(n-1)}]^T$ with presence of uncertainties f, b.

※ To track exactly from $t=0$, $\mathbf{x}_d(0) = \mathbf{x}(0)$.

7.1.1 A notation simplification

$\tilde{x} = x - x_d$ (tracking error), $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$. In the state space $\mathbb{R}^n$,

define $s(\mathbf{x};t) = \left(\frac{d}{dt} + \lambda\right)^{n-1} \tilde{x}$, $S(t) = \{\mathbf{x} \in \mathbb{R}^n : s(\mathbf{x};t) = 0\}$ surface.

strictly positive constant

[(ex) $n=2 \Rightarrow s = \dot{\tilde{x}} + \lambda \tilde{x}$, $n=3 \Rightarrow s = \ddot{\tilde{x}} + 2\lambda \dot{\tilde{x}} + \lambda^2 \tilde{x}$ ...]

$\mathbf{x} \equiv \mathbf{x}_d \Leftrightarrow \begin{cases} s \equiv 0 \\ \mathbf{x}(0) = \mathbf{x}_d(0) \end{cases} \Leftrightarrow \begin{cases} \mathbf{x}(t) \text{ on } S(t) & \forall t \\ \mathbf{x}(0) = \mathbf{x}_d(0) \end{cases}$

- tracking $\mathbf{x}_d$ is n^{th} order tracking problem, but it can be replaced by 1^{st} order stabilization problem in s.
- s represents a true measure of tracking performance in the sense that bound on s gives bounds on $\tilde{\mathbf{x}}$ (At least when $\tilde{\mathbf{x}}(0) = 0$).

$\tilde{x}^{(i)} = \left(\frac{p}{p+\lambda}\right)^i \left(\frac{1}{p+\lambda}\right)^{n-i-1} s$. $\quad y_k = \left(\frac{1}{p+\lambda}\right)^k s$

$z_k = \left(\frac{p}{p+\lambda}\right)^k z_0 \quad (z_0 = y_{n-i-1})$

Note that $y(0) = z(0) = 0$.

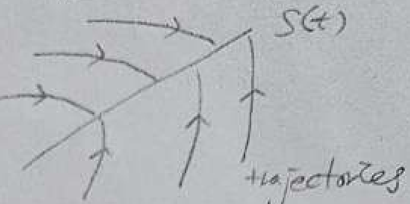
$y_k(t) = e^{-\lambda t} \int_0^t e^{\lambda T} y_{k-1}(T)\, dT \Rightarrow |y_k(t)| \le \|y_{k-1}(t)\|_\infty \cdot \frac{1-e^{-\lambda t}}{\lambda} \le \frac{1}{\lambda} \|y_{k-1}(t)\|_\infty$

Likewise $\le \frac{\|s\|_\infty}{\lambda^k}$

$\|z_k(t)\|_\infty \le \left(1 + \frac{\lambda}{\lambda}\right)^k \|z_1\|_\infty$, $\quad \therefore \|\tilde{x}^{(i)}\|_\infty = \|z_i\|_\infty \le \frac{\|s\|_\infty}{\lambda^{n-i-1}} \cdot 2^i$

$= (2\lambda)^i \varepsilon \quad \left(\varepsilon = \frac{\|s\|_\infty}{\lambda^{n-1}}\right)$

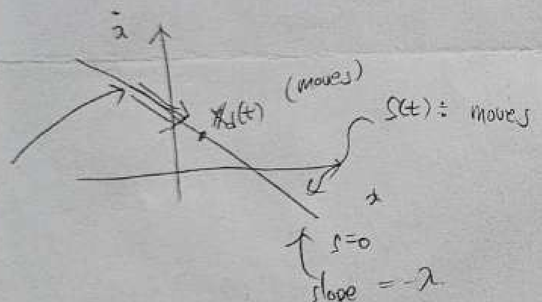
To stay on $S=0$, it's enough to satisfy $\frac{1}{2}\dot{(S^2)} \le -\eta|S|$. (*)
This is called a sliding condition
($S(t)$ invariant set). $S(t)$ with (*) is
referred as a sliding surface, and behavior
on such $S(t)$ is called sliding regime (or mode)



$S=0$ is a dynamics, too.

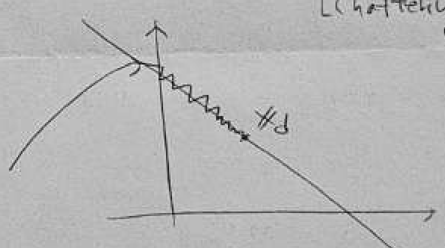
· If (*) is satisfied, $\frac{d}{dt}|S| \le -\eta \rightarrow t_{reach} \le \frac{|S(0)|}{\eta}$.
Once after $S=0$, x converges to x_d exponentially fast.
$\left(\because (D+\lambda)^{n-1}\tilde{x}=0 \rightarrow (D+\lambda)^{n-2}\tilde{x} = c_1 e^{-\lambda t} \cdots \rightarrow \tilde{x} = \left(\sum_{i=0}^{n-2} a_i t^i\right)e^{-\lambda t}\right)$

(Ex)
$n=2 \rightarrow S=0 \Leftrightarrow \dot{\tilde{x}} = -\lambda\tilde{x}$



In Real

[Chattering]



because of discontinuity of u at $S(t)$
→ can't control discontinuously in real life.

In Sec. 7.2 discontinuous u is smoothed.
→ imprecise tracking, but robust control.

7.1.2 Filippov's Construction of the Equivalent Dynamics

(Ex) (7.1.3 basic example) - helps intuition of 7.1.2

$\ddot{x} = f + u$; $|\hat{f} - f| \le F$ (ex) $\left|\begin{array}{l} \ddot{x} + a(t)(\dot{x})^2 \cos 3x = u, \quad 1 \le a(t) \le 2 \\ f = -a(t)(\dot{x})^2 \cos 3x \\ \hat{f} = -1.5(\dot{x})^2 \cos 3x \\ F = 0.5(\dot{x})^2 |\cos 3x| \end{array}\right.$

$s = (p+\lambda)\hat{x} = \dot{\tilde{x}} + \lambda\tilde{x} \Rightarrow \dot{s} = f + u - \ddot{x}_d + \lambda\dot{\tilde{x}}$

$\hat{u}$ for $\dot{s}=0$ is $-\hat{f} + \ddot{x}_d - \lambda\dot{\tilde{x}}$. To satisfy (*),

$u = \hat{u} - k\,\mathrm{sgn}(s) \quad (k(x,\dot{x}))$

$\Rightarrow \frac{1}{2}(\dot{s^2}) = (f-\hat{f})s - k|s|$ let $k = F+\eta \rightarrow \frac{1}{2}(\dot{s^2}) \le -\eta|s|$

Note that $\hat{f}$, F can depend on other (measured) variables or time.

※ $\dot{s} = f - \hat{f} - k\,\mathrm{sgn}(s)$ is not 0 on $S(t)$. But the solution can't escape $S(t)$, since outside $S(t)$ u presses the solution to $S(t)$. Actually, if forcing term ($\dot{y} = f(t,y)$) is discontinuous existence of solution is not guaranteed. In Filippov's book, the weak solution is defined as $\dot{y} \in F(t,y)$ with F is a singleton if f is cont ($F = \{f\}$).

F is taken to approximate well to solutions not considering δ-neighborhood of discontinuity (of f)

There are few ways to define F, but most frequent one is "convex definition"

If $\exists \lim_{s>0} f = f^+$, $\exists \lim_{s<0} f = f^- \Rightarrow F = \mathrm{conv}(f^+, f^-)$

And if $F \cap \text{tangent } S(t) \neq \phi$, let this be f^0.

Solving $\dot{s}=0$ in terms of u gives u_{eq}. This is the same as calculating using "convex definition". Note that $\hat{u}$ of example is u_{eq} with $f \rightarrow \hat{f}$.

equivalent control

7.1.3 Examples

< Integral control >

Formally let $y=\int_0^t \tilde{x}(\tau)\,d\tau$ be a variable of interest.

For previous example, it's third order w.r.t y.

$$s = (D+\lambda)^2 y = \dot{\tilde{x}} + 2\lambda\tilde{x} + \lambda^2\int_0^t \tilde{x}(\tau)\,d\tau$$

$$\dot s = \ddot x - \ddot x_d + 2\lambda\dot{\tilde x} + \lambda^2\tilde x = f + u - \ddot x_d + 2\lambda\dot{\tilde x} + \lambda^2\tilde x$$

$\hat u = -\hat f + \ddot x_d - 2\lambda\dot{\tilde x} - \lambda^2\tilde x$, $u = \hat u - k\,\mathrm{sgn}(s)$ with $k = F+\eta$.

one can use $y = \int_0^t \tilde x + C$ to make $s(0)=0$.

< Gain Margins >

take $\hat b = \sqrt{b_{min} b_{max}}$

$\ddot x = f + bu$; $|f-\hat f| \le F$, $0 < b_{min} \le b \le b_{max}$ $\Rightarrow \beta^{-1} b \le \hat b \le \beta b$

β : gain margin $(\beta > 1)$

$$\dot s = \ddot x - \ddot x_d + \lambda\dot{\tilde x} = f + bu - \ddot x_d + \lambda\dot{\tilde x}$$

$\Rightarrow \hat u = -\hat f + \ddot x_d - \lambda\dot{\tilde x} \Rightarrow u = \hat b^{-1}[\hat u - k\,\mathrm{sgn}(s)]$ with

$k \ge \beta(F+\eta) + (\beta-1)|\hat u|$ satisfies (*) (Note that it increased compared to above)

($\because \dot s = \underbrace{(f - b\hat b^{-1}\hat f) + (1-b\hat b^{-1})(-\ddot x_d + \lambda\dot{\tilde x})} - b\hat b^{-1} k\,\mathrm{sgn}(s)$

To $s\dot s < -\eta|s|$, $b^{-1}\hat b|\quad| + \eta b^{-1}\hat b \le k$ is enough. To remove f,

$f = \hat f + f - \hat f \Rightarrow k \ge |b^{-1}\hat b - 1|\cdot|\hat f - \ddot x_d + \lambda\dot{\tilde x}| + b^{-1}\hat b F + \eta b^{-1}\hat b$)

(Ex) Underwater vehicle

$m\ddot x + c\dot x|\dot x| = u$. $s = \dot{\tilde x} + \lambda\tilde x$. $\dot s = \ddot x - \ddot x_d + \lambda\dot{\tilde x} = \frac{1}{m}(u - c\dot x|\dot x|) - \ddot x_d + \lambda\dot{\tilde x}$

$\hat u = \hat m(\ddot x_d - \lambda\dot{\tilde x}) + \hat c\,\dot x|\dot x|$, $u = \hat u - k\,\mathrm{sgn}\, s$

$$\dot s = \left(\frac{\hat m}{m}-1\right)(\ddot x_d - \lambda\dot{\tilde x}) + \frac{1}{m}\underbrace{(\hat c - c)\dot x|\dot x|}_{\le F} - \frac{k}{m}\mathrm{sgn}\, s$$

$$s\dot s \le \left(\left|\frac{\hat m}{m}-1\right||\ddot x_d - \lambda\dot{\tilde x}| + \frac{F}{m} - \frac{k}{m}\right)|s|$$

$\rightarrow k \ge F + \eta\cdot m + |m-\hat m|_{max}|\ddot x_d - \lambda\dot{\tilde x}| \rightarrow k = F + \hat m(\beta\eta + |\beta-1|\,|\ddot x_d - \lambda\dot{\tilde x}|)$

For given feasible trajectory x_d, letting $u(t) = b(x_d)^{-1}[\ddot{x}_d - f(x_d)]$ gives

$\ddot{x} = \ddot{x}_d + f(x) - f(x_d) \rightarrow x = x_d$ is a solution. If u cont, it's unique solution.

In the presence of uncertainties, $u(t)$ becomes discontinuous to achieve perfect tracking → in practice chattering appears.

In general, chattering is bad, but is applicable in some situations

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7.1.4 Direct Implementations of Switching Control Laws

<Switching Control in place of Pulse-Width modulation>

Pulse-width modulated electric motor : Controling DC motors with electrical pulses.

⇒ Unlike the control by mechanical force, very high frequency, accurate control is possible, so sliding control gives extremely high performance.

<Switching Control with Linear observer>

Measuring state accurately with high frequency is hard ⇒ use state observer (state estimator)

(For linear case, building state observer is easy)

One can make $x_e \rightarrow x$ observer; so control via $s_e = 0$ (using x_e)

<Switching control in place of Dither>

Static friction, stiction, hysteresis (이력) cause uncertainties

→ Dither, small ripple frequency can overcome these effects.

Instead of adding ripple to input, using switching control have similar effect.

In the next chapter, we focus to reduce chattering.

$$\cdot\dot{X}\cdot \left[\begin{array}{l} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) + Du(k) \end{array}\right. \xrightarrow{\text{state observer}} \left[\begin{array}{l} \hat{x}(k+1) = A\hat{x}(k) + L[y(k) - \hat{y}(k)] + Bu(k) \\ \hat{y}(k) = C\hat{x}(k) + Du(k) \end{array}\right. \quad \boxed{\lambda_{max}(A - LC) < 1}$$